Andrea González

dreagon2010@gmail.com | linkedin.com/in/dreagonzalezz | New York, NY | Portfolio

EDUCATION

New York University **May 2026**
Bachelor of Science in Computer Engineering *New York, NY*

EXPERIENCE

Medtronic **Jun 2026 – Aug 2026**
Software Engineering Intern *Mounds View, MN*

- Developed an AI agent that matches medical device complaints to hazard codes, cutting matching time by ∼95%
- Enabled full traceability from complaints to hazard codes, strengthening risk defensibility for regulatory audits
- Built a 40-code hazard library from 400+ hazard logs, validated with reliability and post-market engineers
- Analyzed agent output to identify gaps in hazard code coverage and proactively generated missing hazard codes

NASA Lunabotics **Jun 2025 – May 2026**
Lead Software Engineer *New York, NY*

- Built a Gazebo simulation to test algorithms before hardware runs, reducing physical testing time by ∼40%
- Managed sprint planning and backlog via GitHub Projects using Agile methodology, coordinating work across team
- Designed a multi-node ROS architecture over TCPROS/UDPROS Wi-Fi, enabling remote perception and control

NYU Financial Operations & Treasury Division **Apr 2024 – May 2025**
E-commerce Assistant *New York, NY*

- Built 30+ eCommerce UStores with TouchNet to improve UX and purchasing flows for campus organizations
- Validated test environments with clients and managed production deployments, ensuring reliable launches

PROJECTS

Rover Autonomous Navigation | *ROS2, Python, Ubuntu, Gazebo, RTAB-Map, Nav2*

- Integrated encoder, load cell, and camera data to drive state machines for rover's autonomous decision logic
- Built real-time 3D terrain maps using RTAB-Map, enabling reliable obstacle detection on unstructured terrain
- Deployed Nav2 for path planning and obstacle avoidance, cutting path-finding time by ∼20% in competition

Stanford Pupper v3 Object Tracking | *ROS2, Python, OpenCV*

- Developed a real-time object tracking and following system on a quadruped robot, integrating custom ROS2 perception nodes with a neural locomotion controller
- Built an OpenCV-based detection pipeline that reliably identified and followed a target object, closing the loop between vision and physical motion

Visual Navigation Game | *PyTorch, CNN, Python*

- Built a vision-based navigation pipeline in PyTorch, tuning a CNN with SGD and regularization to improve goal-detection accuracy in a simulated robotic environment

LEADERSHIP & PROFESSIONAL DEVELOPMENT

NYU Society of Women Engineers **Jan 2026 – May 2026**
Vice President *New York, NY*

Finnegan, Henderson, Farabow, Garrett & Dunner LLP **Feb 2026**
Finnegan IP University Participant *Remote*

Google Developer Groups **Oct 2024 – Dec 2024**
Frontend Developer *New York, NY*

SKILLS

Languages: Python, C++, SQL, MATLAB
Robotics & Data: ROS2, Gazebo, Nav2, RTAB-Map, OpenCV, PyTorch
Tools & Hardware: Git, Linux (Ubuntu), Flask, Jetson, Raspberry Pi, Arduino